

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – Scenario-Based

Course Code:	CSA5802	Course Title:	DEVSECOPS ENGINEERING AND APPLICATION SECURITY
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2
Weightage:	30% of CIA	Total Marks:	100
CO1 Statement:	Team-Based Concept Association, Knowledge Competition and Collaborative Learning.		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Instructions to Students

- Each team receives one unique industry case study.
- Analyze the scenario and complete all investigation tasks.
- Prepare a 5-minute presentation and one consolidated report.
- Support every recommendation with technical justification.

Question Type	Focus Area	Questions	Marks
Concept Mapping	Scenario-Based Collaborative Analysis	Q1	Q1 –100
GRAND TOTAL:		100 Marks	

SECTION A

Team 1 – Digital Banking Platform

Background

SecureBank Ltd. operates a nationwide digital banking platform serving more than five million customers. The bank offers online banking, mobile banking, digital payments, loan processing and investment services. To remain competitive with fintech companies, management mandates weekly software releases.

The development team commits directly to the main branch, code reviews are optional, security testing is performed only before production deployment, third-party libraries are adopted without verification, password policies are weak, and API authentication is inconsistent. Application logs are generated but rarely reviewed. Developers have not received secure coding training.

Recently, customers reported unauthorized login attempts and suspicious transactions. Internal investigation revealed exposed API keys, delayed security patches and outdated libraries.

Investigation Questions

1. Identify at least five security risks.
2. Identify at least five vulnerabilities and classify each as High, Medium or Low.
3. List possible cyber threats and explain their impact.
4. Prepare a DevOps vs DevSecOps comparison table (minimum eight points).
5. Recommend five OWASP security best practices.
6. Explain the business impact if these issues remain unresolved.
7. Create a concept map linking Risk → Vulnerability → DevSecOps Practice → Business Benefit.
8. Prepare a five-minute presentation summarizing your findings.

Team 2 – E-Commerce Platform

Background

ShopNow is a fast-growing online marketplace processing thousands of customer orders every day. Marketing campaigns require rapid feature delivery, so developers frequently deploy directly to production. Peer reviews are skipped, secure coding standards are not enforced, payment services are integrated with multiple third-party vendors, and vulnerability assessments are performed only after incidents.

Customers have recently complained about fraudulent purchases and unexpected account activity. Management suspects weaknesses in authentication, session handling and application security.

Investigation Questions

1. Identify at least five security risks.
2. Identify at least five vulnerabilities and classify each as High, Medium or Low.
3. List possible cyber threats and explain their impact.
4. Prepare a DevOps vs DevSecOps comparison table (minimum eight points).
5. Recommend five OWASP security best practices.
6. Explain the business impact if these issues remain unresolved.
7. Create a concept map linking Risk → Vulnerability → DevSecOps Practice → Business Benefit.
8. Prepare a five-minute presentation summarizing your findings.

Team 3 – Hospital Information System

Background

CarePlus Hospital provides an online patient portal containing electronic medical records, prescriptions, laboratory reports and appointment scheduling. Weekly updates are released to improve patient experience. Passwords are stored using weak practices, security reviews are skipped to meet deadlines, and staff access the system from unmanaged devices.

A recent audit found unauthorized access attempts against patient records. Hospital management is concerned about patient privacy, regulatory compliance and service availability.

Investigation Questions

1. Identify at least five security risks.
2. Identify at least five vulnerabilities and classify each as High, Medium or Low.
3. List possible cyber threats and explain their impact.
4. Prepare a DevOps vs DevSecOps comparison table (minimum eight points).
5. Recommend five OWASP security best practices.
6. Explain the business impact if these issues remain unresolved.
7. Create a concept map linking Risk → Vulnerability → DevSecOps Practice → Business Benefit.
8. Prepare a five-minute presentation summarizing your findings.

Team 4 – University ERP Portal

Background

A university ERP system manages admissions, examinations, attendance, fee payments and student records for over 20,000 students. Development focuses primarily on new functionality. Password reset links never expire, uploaded files are not validated, security testing occurs only after deployment and user activity logs are rarely reviewed.

The administration recently discovered unusual changes in student records and suspects weaknesses in access control and authentication.

Investigation Questions

1. Identify at least five security risks.
2. Identify at least five vulnerabilities and classify each as High, Medium or Low.
3. List possible cyber threats and explain their impact.
4. Prepare a DevOps vs DevSecOps comparison table (minimum eight points).
5. Recommend five OWASP security best practices.
6. Explain the business impact if these issues remain unresolved.
7. Create a concept map linking Risk → Vulnerability → DevSecOps Practice → Business Benefit.
8. Prepare a five-minute presentation summarizing your findings.

Team 5 – Airline Reservation System

Background

SkyFly Airlines continuously updates its online booking platform with promotional offers and new services. The application depends heavily on open-source libraries but no formal process exists for checking known vulnerabilities. Patches are delayed to avoid interrupting business operations, API protection is inconsistent and security reviews are reactive.

A recent outage disrupted online ticket bookings, causing revenue loss and customer complaints.

Investigation Questions

1. Identify at least five security risks.
2. Identify at least five vulnerabilities and classify each as High, Medium or Low.
3. List possible cyber threats and explain their impact.
4. Prepare a DevOps vs DevSecOps comparison table (minimum eight points).
5. Recommend five OWASP security best practices.
6. Explain the business impact if these issues remain unresolved.
7. Create a concept map linking Risk → Vulnerability → DevSecOps Practice → Business Benefit.
8. Prepare a five-minute presentation summarizing your findings.

Team 6 – Government Citizen Services Portal

Background

A national citizen services portal provides tax filing, identity services, certificate applications and public grievance management. Different development teams build independent modules using inconsistent security practices. Identity management is weak, security policies are documented but rarely enforced, logging is limited and monitoring is largely manual.

An external audit highlighted several compliance concerns and recommended immediate security improvements.

Investigation Questions

1. Identify at least five security risks.
2. Identify at least five vulnerabilities and classify each as High, Medium or Low.
3. List possible cyber threats and explain their impact.
4. Prepare a DevOps vs DevSecOps comparison table (minimum eight points).
5. Recommend five OWASP security best practices.
6. Explain the business impact if these issues remain unresolved.
7. Create a concept map linking Risk → Vulnerability → DevSecOps Practice → Business Benefit.
8. Prepare a five-minute presentation summarizing your findings.

Team 7 – Social Media Platform

Background

ConnectHub is a rapidly expanding social networking platform supporting millions of users who upload photos, videos and messages daily. The platform exposes multiple public APIs to mobile applications and third-party partners. Access controls differ across services, suspicious user behaviour is rarely monitored and the security team is involved only after incidents.

Recent reports indicate fake accounts, unauthorized data access and leaked user information, raising concerns about privacy and trust.

Investigation Questions

1. Identify at least five security risks.
2. Identify at least five vulnerabilities and classify each as High, Medium or Low.
3. List possible cyber threats and explain their impact.
4. Prepare a DevOps vs DevSecOps comparison table (minimum eight points).
5. Recommend five OWASP security best practices.
6. Explain the business impact if these issues remain unresolved.
7. Create a concept map linking Risk → Vulnerability → DevSecOps Practice → Business Benefit.
8. Prepare a five-minute presentation summarizing your findings.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Rubrics:

Criteria	Weightage	Awarded Marks	Total Marks (100)
Understanding of Concepts	25%		
Experimental Design	30%		
Use of Tools	20%		
Analysis of Results	15%		
Documentation	10%		
Total per Question	100 Marks		

100

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____